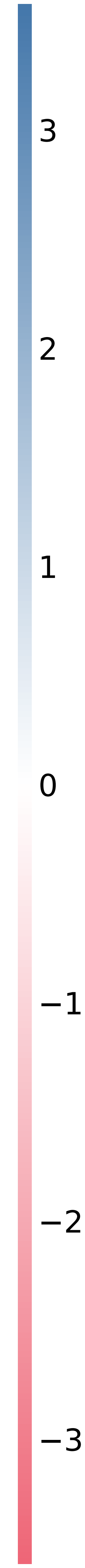
Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal
B. 40 Kcal, 5 Kcal
C. 30 Kcal, 8.58 Kcal
D. 25 Kcal, 8 Kcal
E. 29 Kcal, 7.77 Kcal
F. 34 Kcal, 6.9 Kcal
G. 30 Kcal, 6.45 Kcal
H. 32 Kcal, 7 Kcal
I. 35 Kcal, 7.5 Kcal
J. 31 Kcal, 6.15 Kcal



Reasoning + Answer:

<think>
To solve this problem, we need to use the formula for heat transfer by a substance due to temperature change: Q = mcÎĶT. Here, Q is the heat in cal, m is the mass in g, c is the specific heat, and ÎĶT is the change in temperature in Â°C.

(a) For water:
m = 1 liter * 1 g/ml * 1000 ml/liter = 1000 g
c = 1 cal/gÂ°C
ÎĶT = 60 - 30 = 30Â°C

Q = mcÎĶT = 1000 g * 1 cal/gÂ°C * 30Â°C = 30000 cal = 30 Kcal.

(b) For aluminum:
m = 1 kg * 1000 g/kg = 1000 g
c = 0.215 cal/gÂ°C
ÎĶT = 60 - 30 = 30Â°C

Q = mcÎĶT = 1000 g * 0.215 cal/gÂ°C * 30Â°C = 6435 cal = 6.435 Kcal, which rounds to 6.45 Kcal.
</think><answer>G.